

|                                                                                                                         |                             |                   |
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|  <b>BORDEAUX<br/>SCIENCES<br/>AGRO</b> | <b>Clearpass</b>            | <b>04/03/2025</b> |
|                                                                                                                         | <b>Notes d'installation</b> |                   |

# Clearpass

## Notes d'installation

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## 1 Reste à faire

- Portail captif filaire
- Libre service
- Switch Legacy (1910,1930,2530-8)
- TACACS

Questions :

- Authentification sur Bxsa\_DEVICE : juste l'uid ?
- 

## 2 Clés radius

CLE RADIUS CLEARPASS / SWITCHS : !PD@sdhV43\_zb48\_g\$8gTRicH&n\*3V/  
 CLE RADIUS CLEARPASS / BORNES : t9YSGA75\_=-UvbaZw3j5TNquR8d.\_\*%

## 3 Licences Clearpass

Enregistrement chez HPE :

<https://networkingsupport.hpe.com/home>

License Management System

## 4 Notes

Sur POMS-ST-85, ne pas faire de trunk all à cause des VLANs WAN (602, 1711)

Avec clearpass, plus besoin de mettre les VLANs sur les ports de distribution. Mettre les ports d'interco en trunk all. Sauf pour les 3 switchs 2530-89780A

Les noms des VLANs sur les switchs sont importants et doivent respecter ceux entrés dans Clearpass

## 5 Configuration postes clients filaires

### 5.1 Windows 10

Vérifier que le service « Configuration automatique de réseau câblé »

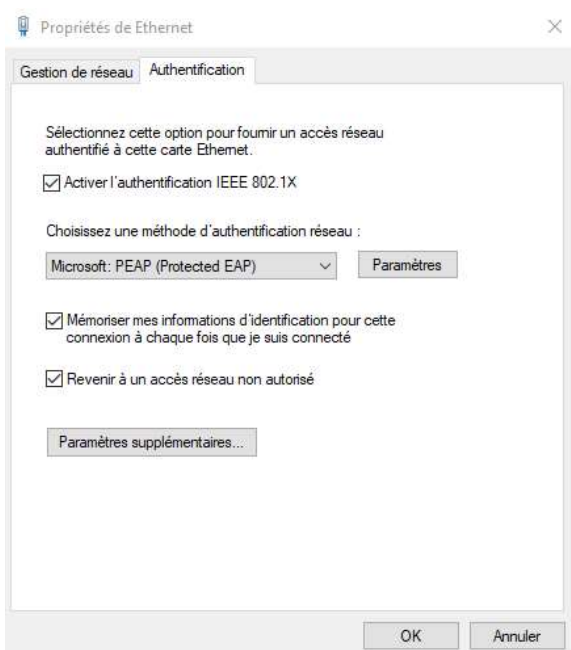
Paramètres / Réseau et internet / Ethernet

Propriété de la carte réseau

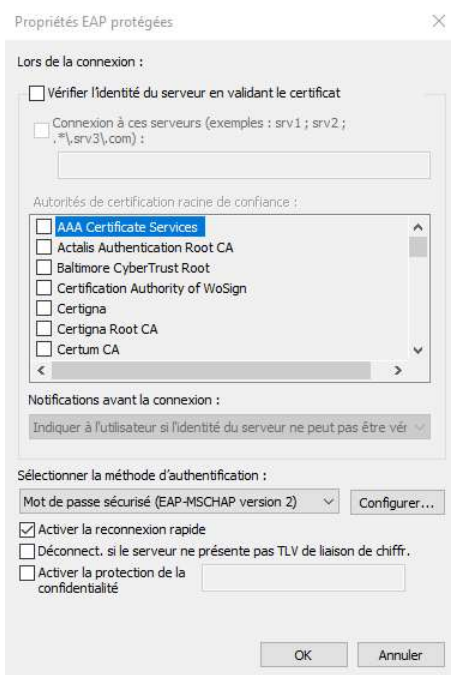
Onglet Authentification

Paramètres

|                                                                                                                         |                             |                   |
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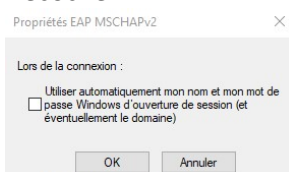


Décocher « Vérifier l'identité du serveur »



Configurer

Décocher



|                                                                                                                         |                             |                   |
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## 5.2 Windows 11

Paramètres / Réseau et internet / Ethernet

Déplier les propriétés de la carte

Paramètres d'authentification : Modifier

Activer le 802.1X

Modifier la configuration

Méthode EAP : PEP

Méthode d'authentification : Mot de passe sécurisé (EAP-MSCHAP version 2)

Enregistrer

## 6 Configuration Switchs

### 6.1 Switchs 2530 en YB (J9780A)

Exclure les ports où sont connectées les bornes

```
encrypt-credentials
```

Répondre Y

```
radius-server host 192.168.70.100 encrypted-key
"giU8jb1c1R+O8gNEJtAtUcv2qQkgRs24EcUYKwM79fyLFqsMH6T95qALzRw6sJqG41l/wrvW/ZEvFBZokXCIDQ=="
radius-server host 192.168.70.100 dyn-authorization
radius-server host 192.168.70.100 time-window plus-or-minus-time-window
radius-server host 192.168.70.100 time-window 30
ip client-tracker trusted
aaa server-group radius "CLEARPASS" host 192.168.70.100
aaa accounting update periodic 5
aaa accounting network start-stop radius server-group "CLEARPASS"

aaa authentication port-access eap-radius server-group "CLEARPASS" cached-reauth
```

Répondre Y

```
aaa authentication mac-based chap-radius server-group "CLEARPASS"
aaa authentication captive-portal enable
```

```
aaa port-access authenticator 1-4
aaa port-access authenticator 1-4 tx-period 10
aaa port-access authenticator 1-4 supplicant-timeout 10
aaa port-access authenticator 1-4 client-limit 8
aaa port-access authenticator active
aaa port-access 1-4 auth-order authenticator mac-based
```

```
aaa port-access mac-based 1-4
aaa port-access mac-based 1-4 addr-limit 8
```

```
spanning-tree
spanning-tree 1-4 admin-edge-port
spanning-tree 1-4 bpdu-filter
spanning-tree mode rapid-pvst
```

Répondre Y

|                                                                                                                         |                             |                   |
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## 6.2 switch 2540 et 2530 en YA

```

class ipv4 "DNS"
  10 match udp 0.0.0.0 255.255.255.255 0.0.0.0 255.255.255.255 eq 53
  exit
class ipv4 "DHCP"
  10 match udp 0.0.0.0 255.255.255.255 0.0.0.0 255.255.255.255 eq 67
  exit
class ipv4 "WEB-TRAFFIC"
  10 match tcp 0.0.0.0 255.255.255.255 0.0.0.0 255.255.255.255 eq 80
  20 match tcp 0.0.0.0 255.255.255.255 0.0.0.0 255.255.255.255 eq 443
  exit
class ipv4 "CLEARPASS-WEB"
  10 match tcp 0.0.0.0 255.255.255.255 10.1.254.240 0.0.0.0 eq 80
  20 match tcp 0.0.0.0 255.255.255.255 10.1.254.240 0.0.0.0 eq 443
  exit
policy user "CP-CLEARPASS-REDIRECT"
  10 class ipv4 "DNS" action permit
  20 class ipv4 "DHCP" action permit
  30 class ipv4 "CLEARPASS-WEB" action permit
  40 class ipv4 "WEB-TRAFFIC" action redirect captive-portal
  exit
encrypt-credentials

```

### Répondre Y

```

radius-server host 192.168.70.100 encrypted-key
"giU8jb1c1R+O8gNEJtAtUcv2qQkgrs24EcUYKwM79fyLFqsMH6T95qALzRw6sJqG4ll/wrvW/ZEvFBZokXCIDQ=="
radius-server host 192.168.70.100 dyn-authorization
radius-server host 192.168.70.100 time-window plus-or-minus-time-window
radius-server host 192.168.70.100 time-window 30
ip client-tracker trusted
aaa server-group radius "CLEARPASS" host 192.168.70.100
aaa accounting update periodic 5
aaa accounting network start-stop radius server-group "CLEARPASS"
aaa authorization user-role name "LUR-IOT"
  reauth-period 10800
  vlan-id 81
  exit
aaa authorization user-role name "LUR-TOIP"
  reauth-period 10800
  vlan-id-tagged 20
  exit
aaa authorization user-role name "LUR-AGENTS"
  reauth-period 10800
  vlan-id 78
  exit
aaa authorization user-role name "LUR-BORNES"
  reauth-period 10800
  vlan-id 80
  vlan-id-tagged 4,77-79,81-83
  device
    port-mode
  exit
  exit
aaa authorization user-role name "LUR-FLUIDES"
  reauth-period 10800
  vlan-id 6
  exit
aaa authorization user-role name "LUR-COPIEURS"
  reauth-period 10800
  vlan-id 3
  exit
aaa authorization user-role name "LUR-HEBERGES"
  reauth-period 10800
  vlan-id 83
  exit
aaa authorization user-role name "LUR-APPRENANTS"
  reauth-period 10800

```

|                                                                                                                         |                             |                   |
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```

vlan-id 79
exit
aaa authorization user-role name "LUR-AUDIOVISUEL"
reauth-period 10800
vlan-id 4
exit
aaa authorization user-role name "LUR-LIBRESERVICE"
reauth-period 10800
vlan-id 2
exit
aaa authorization user-role name "LUR-CONTROLEACCES"
reauth-period 10800
vlan-id 5
exit
aaa authorization user-role name "LUR-GUESTS-PREAUTH"
captive-portal-profile "CP-Clearpass"
policy "CP-CLEARPASS-REDIRECT"
vlan-id 82
exit
aaa authorization user-role name "LUR-GUESTS-PROFILING"
captive-portal-profile "CP-Clearpass"
policy "CP-CLEARPASS-REDIRECT"
reauth-period 180
vlan-id 82
exit
aaa authorization user-role enable
aaa authentication port-access eap-radius server-group "CLEARPASS" cached-reauth
aaa authentication mac-based chap-radius server-group "CLEARPASS"
aaa authentication captive-portal enable
aaa authentication captive-portal profile "CP-Clearpass" url https://portail.agro-bordeaux.fr/visiteurs\_filaire\_legacy\_portal.php

```

## 6.3 switch 2540

```

class ipv4 "DNS"
  10 match udp 0.0.0.0 255.255.255.255 0.0.0.0 255.255.255.255 eq 53
  exit
class ipv4 "DHCP"
  10 match udp 0.0.0.0 255.255.255.255 0.0.0.0 255.255.255.255 eq 67
  exit
class ipv4 "WEB-TRAFFIC"
  10 match tcp 0.0.0.0 255.255.255.255 0.0.0.0 255.255.255.255 eq 80
  20 match tcp 0.0.0.0 255.255.255.255 0.0.0.0 255.255.255.255 eq 443
  exit
class ipv4 "CLEARPASS-WEB"
  10 match tcp 0.0.0.0 255.255.255.255 10.1.254.240 0.0.0.0 eq 80
  20 match tcp 0.0.0.0 255.255.255.255 10.1.254.240 0.0.0.0 eq 443
  exit
policy user "CP-CLEARPASS-REDIRECT"
  10 class ipv4 "DNS" action permit
  20 class ipv4 "DHCP" action permit
  30 class ipv4 "CLEARPASS-WEB" action permit
  40 class ipv4 "WEB-TRAFFIC" action redirect captive-portal
  exit
encrypt-credentials
radius-server host 192.168.70.100 encrypted-key
"giU8jb1c1R+08gNEJtAtUcv2qQkgrs24EcUYKwM79fyLFqsMH6T95qALzRw6sJqG4ll/wrvW/ZEvFBZokXCIDQ=="
radius-server host 192.168.70.100 dyn-authorization
radius-server host 192.168.70.100 time-window plus-or-minus-time-window
radius-server host 192.168.70.100 time-window 30
ip client-tracker trusted
aaa server-group radius "CLEARPASS" host 192.168.70.100
aaa accounting update periodic 5
aaa accounting network start-stop radius server-group "CLEARPASS"
aaa authorization user-role name "LUR-IOT"
reauth-period 10800
vlan-id 81
exit

```

|                                                                                                                         |                             |                   |
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```

aaa authorization user-role name "LUR-TOIP"
    reauth-period 10800
    vlan-id-tagged 20
    exit
aaa authorization user-role name "LUR-AGENTS"
    reauth-period 10800
    vlan-id 78
    exit
aaa authorization user-role name "LUR-BORNES"
    reauth-period 10800
    vlan-id 80
    vlan-id-tagged 4,77-79,81-83
    device
        port-mode
    exit
    exit
aaa authorization user-role name "LUR-FLUIDES"
    reauth-period 10800
    vlan-id 6
    exit
aaa authorization user-role name "LUR-COPIEURS"
    reauth-period 10800
    vlan-id 3
    exit
aaa authorization user-role name "LUR-HEBERGES"
    reauth-period 10800
    vlan-id 83
    exit
aaa authorization user-role name "LUR-APPRENANTS"
    reauth-period 10800
    vlan-id 79
    exit
aaa authorization user-role name "LUR-AUDIOVISUEL"
    reauth-period 10800
    vlan-id 4
    exit
aaa authorization user-role name "LUR-LIBRESERVICE"
    reauth-period 10800
    vlan-id 2
    exit
aaa authorization user-role name "LUR-CONTROLEACCES"
    reauth-period 10800
    vlan-id 5
    exit
aaa authorization user-role name "LUR-GUESTS-PREAUTH"
    captive-portal-profile "CP-Clearpass"
    policy "CP-CLEARPASS-REDIRECT"
    vlan-id 82
    exit
aaa authorization user-role name "LUR-GUESTS-PROFILING"
    captive-portal-profile "CP-Clearpass"
    policy "CP-CLEARPASS-REDIRECT"
    reauth-period 180
    vlan-id 82
    exit
aaa authorization user-role enable
aaa authentication port-access eap-radius server-group "CLEARPASS" cached-reauth
aaa authentication mac-based chap-radius server-group "CLEARPASS"
aaa authentication captive-portal enable
aaa authentication captive-portal profile "CP-Clearpass" url "https://portail.agro-
bordeaux.fr/visiteurs_filaire_legacy_portal.php"
aaa port-access authenticator 1-4
aaa port-access authenticator 1-4 tx-period 10
aaa port-access authenticator 1-4 supplicant-timeout 10
aaa port-access authenticator 1-4 client-limit 8
aaa port-access authenticator active
aaa port-access 1 auth-order authenticator mac-based

Configuration des ports pour l'authentification par MACADDRESS
aaa port-access mac-based 1-4
aaa port-access mac-based 1-4 addr-limit 8

```

|                                                                                                                         |                             |                   |
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```
spanning-tree
spanning-tree 1-4 admin-edge-port
spanning-tree 1-4 bpdu-filter
spanning-tree mode rapid-pvst
```

## 6.4 Explications

### Redirection Portail Captif

```
class ipv4 "DNS"
    10 match udp 0.0.0.0 255.255.255.255 0.0.0.0 255.255.255.255 eq 53
    exit
class ipv4 "DHCP"
    10 match udp 0.0.0.0 255.255.255.255 0.0.0.0 255.255.255.255 eq 67
    exit
class ipv4 "WEB-TRAFFIC"
    10 match tcp 0.0.0.0 255.255.255.255 0.0.0.0 255.255.255.255 eq 80
    20 match tcp 0.0.0.0 255.255.255.255 0.0.0.0 255.255.255.255 eq 443
    exit
class ipv4 "CLEARPASS-WEB"
    10 match tcp 0.0.0.0 255.255.255.255 10.1.254.240 0.0.0.0 eq 80
    20 match tcp 0.0.0.0 255.255.255.255 10.1.254.240 0.0.0.0 eq 443
    exit
policy user "CP-CLEARPASS-REDIRECT"
    10 class ipv4 "DNS" action permit
    20 class ipv4 "DHCP" action permit
    30 class ipv4 "CLEARPASS-WEB" action permit
    40 class ipv4 "WEB-TRAFFIC" action redirect captive-portal
Exit
```

### Chiffrer les mots de passe (radius, admin, etc)

```
encrypt-credentials
```

### Serveur Radius

```
radius-server host 192.168.70.100 encrypted-key
"giU8jb1c1R+O8gNEJtAtUcv2qQkgrs24EcUYKwM79fyLFqsMH6T95qALzRw6sJqG41l/wrvW/ZEvFBZokXCIDQ=="
```

### Autorisation Changement de vlan

```
radius-server host 192.168.70.100 dyn-authorization
radius-server host 192.168.70.100 time-window plus-or-minus-time-window
radius-server host 192.168.70.100 time-window 30
```

### Pour suivre les IPs

```
ip client-tracker trusted
```

### Envoi des logs au serveur Clearpass

```
aaa server-group radius "CLEARPASS" host 192.168.70.100
aaa accounting update periodic 5
aaa accounting network start-stop radius server-group "CLEARPASS"
```

### Définition des rôles et des vlans associés

(vlan-id = untagged, vlan-id-tagged = tagged)

```
aaa authorization user-role name "LUR-IOT"
    reauth-period 10800
    vlan-id 81
    exit
aaa authorization user-role name "LUR-TOIP"
    reauth-period 10800
    vlan-id-tagged 20
    exit
aaa authorization user-role name "LUR-AGENTS"
    reauth-period 10800
    vlan-id 78
    exit
aaa authorization user-role name "LUR-BORNES"
```



|                                                                                                                         |                             |                   |
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```

reauth-period 10800
vlan-id 80
vlan-id-tagged 4,77-79,81-83
device
    port-mode
    exit
exit
aaa authorization user-role name "LUR-FLUIDES"
reauth-period 10800
vlan-id 6
exit
aaa authorization user-role name "LUR-COPIEURS"
reauth-period 10800
vlan-id 3
exit
aaa authorization user-role name "LUR-HEBERGES"
reauth-period 10800
vlan-id 83
exit
aaa authorization user-role name "LUR-APPRENANTS"
reauth-period 10800
vlan-id 79
exit
aaa authorization user-role name "LUR-AUDIOVISUEL"
reauth-period 10800
vlan-id 4
exit
aaa authorization user-role name "LUR-LIBRESERVICE"
reauth-period 10800
vlan-id 2
exit
aaa authorization user-role name "LUR-CONTROLEACCES"
reauth-period 10800
vlan-id 5
exit
aaa authorization user-role name "LUR-GUESTS-PREAUTH"
captive-portal-profile "CP-Clearpass"
policy "CP-CLEARPASS-REDIRECT"
vlan-id 82
exit
aaa authorization user-role name "LUR-GUESTS-PROFILING"
captive-portal-profile "CP-Clearpass"
policy "CP-CLEARPASS-REDIRECT"
reauth-period 180
vlan-id 82
exit

```

Autorise le mapping des roles du clearpass et des rôles locaux

```
aaa authorization user-role enable
```

Active le 802.1X

```
aaa authentication port-access eap-radius server-group "CLEARPASS" cached-reauth
```

Active l'authentification par MAC-ADDRESS

```
aaa authentication mac-based chap-radius server-group "CLEARPASS"
```

Active le portail captif

```
aaa authentication captive-portal enable
```

```
aaa authentication captive-portal profile "CP-Clearpass" url https://portail.agro-bordeaux.fr/visiteurs\_filaire\_legacy\_portal.php
```

Configuration des ports pour 802.1X

```
aaa port-access authenticator 1-4
```

```
aaa port-access authenticator 1-4 tx-period 10
```

```
aaa port-access authenticator 1-4 supplicant-timeout 10
```

```
aaa port-access authenticator 1-4 client-limit 8
```

```
aaa port-access authenticator active
```

Priorisation de l'authentification par 802.1X sur MAC based

```
aaa port-access 1 auth-order authenticator mac-based
```

|                                                                                                                         |                             |                   |
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Configuration des ports pour l'authentification par MACADDRESS

```
aaa port-access mac-based 1-4
aaa port-access mac-based 1-4 addr-limit 8
```

Modification du comportement du spanning-tree sur les ports de distribution

Ne pas faire les ports d'interco !!!

```
spanning-tree
spanning-tree 1-4 admin-edge-port
spanning-tree 1-4 bpdu-filter
spanning-tree mode rapid-pvst
```

## 7 Ajout d'un switch dans Clearpass

Policy Manager

Configuration

Network

Devices

Copier un switch équivalent

Renseigner Name, IP

Save

Cliquer le switch

Renseigner Description, Vendor name

Normalement la clé est copiée

Save

Device Groups

Cliquer sur le groupe du switch :

BSA – SWITCH – AOS : 2530 (sauf 2530-8), 2540

BSA – SWITCH – AOS-CX : 6X00

BSA – SWITCH – LEGACY tous les autres (1910, 1930, 2530-8)

Ajouter le switch dans le groupe

Save

## 8 Portail Captif

### 8.1 Personnalisation

<https://clearpass1.agro-bordeaux.fr/tips/welcome.action>

ClearPass guest

Administration

Gestionnaire de plugin

Galleria Skin

Configuration

Pour uploader les images, logos :

Configuration

Gestionnaire de contenu

|                                                                                                                        |                             |            |
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Pour vérifier : [https://portail.agro-bordeaux.fr/guest/enregistrement\\_visiteurs\\_login.php](https://portail.agro-bordeaux.fr/guest/enregistrement_visiteurs_login.php)

## 8.2 Envoi de SMS

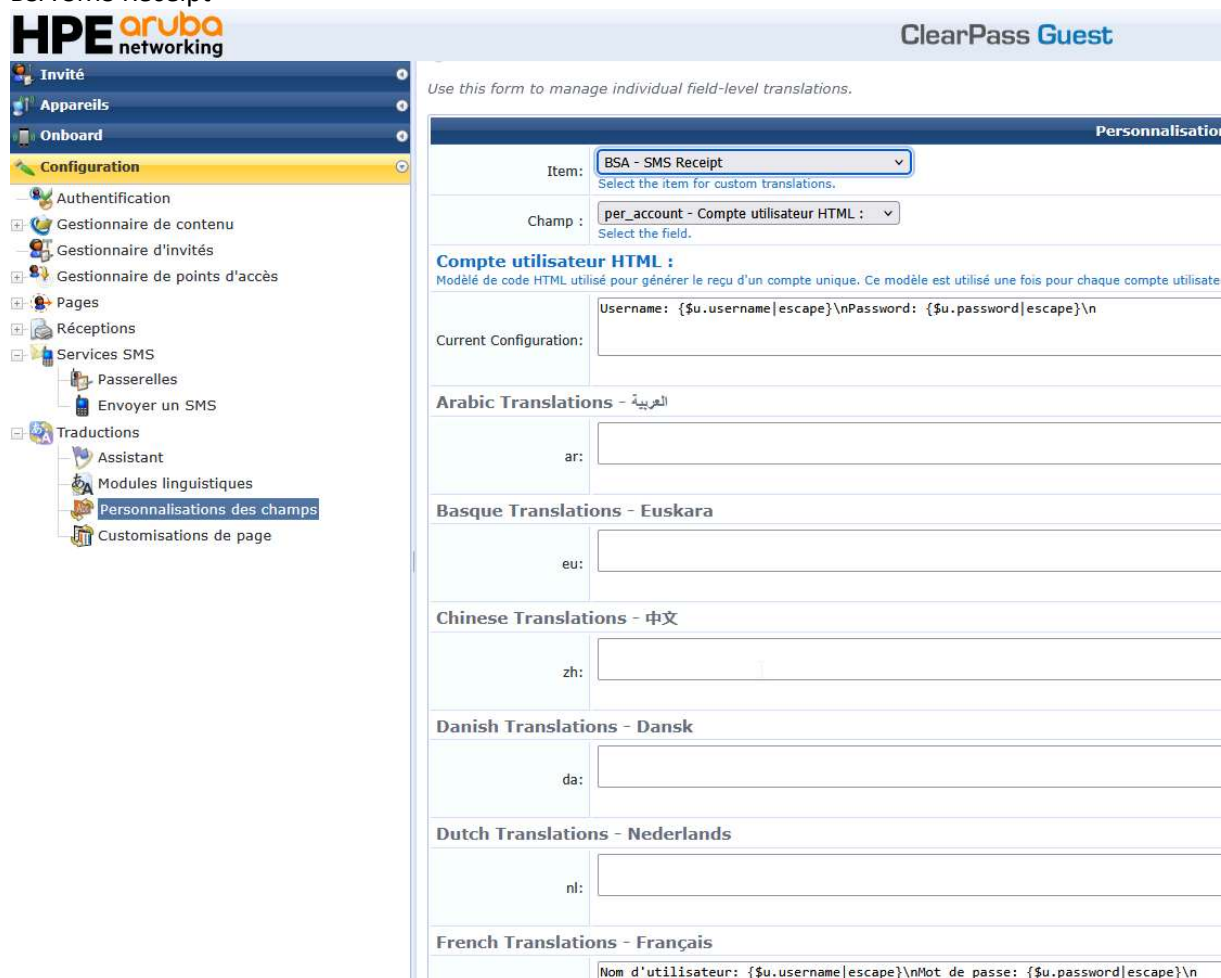
Fournisseur SMSEnvoi

SMSEnvoi veut des \n à la place des retours chariots

Il faut modifier les champs dans

ClearPass Guest / Traductions / Personnalisation des champs

BSA SMS Receipt



The screenshot shows the 'ClearPass Guest' configuration page for 'BSA - SMS Receipt'. The left sidebar contains a navigation menu with options like 'Invité', 'Appareils', 'Onboard', 'Configuration', 'Authentification', 'Gestionnaire de contenu', 'Gestionnaire d'invités', 'Gestionnaire de points d'accès', 'Pages', 'Réceptions', 'Services SMS', 'Passerelles', 'Envoyer un SMS', 'Traductions', 'Assistant', 'Modules linguistiques', 'Personnalisations des champs', and 'Customisations de page'. The main area is titled 'Personnalisation' and includes a description: 'Use this form to manage individual field-level translations.' It features a dropdown for 'Item' (BSA - SMS Receipt) and a dropdown for 'Champ' (per\_account - Compte utilisateur HTML). Below this, the 'Compte utilisateur HTML' section shows the current configuration: 'Username: { \$u.username|escape }\nPassword: { \$u.password|escape }\n'. The 'Arabic Translations - العربية' section shows the 'ar:' field. The 'Basque Translations - Euskara' section shows the 'eu:' field. The 'Chinese Translations - 中文' section shows the 'zh:' field. The 'Danish Translations - Dansk' section shows the 'da:' field. The 'Dutch Translations - Nederlands' section shows the 'nl:' field. The 'French Translations - Français' section shows the 'Nom d'utilisateur: { \$u.username|escape }\nMot de passe: { \$u.password|escape }\n'.

## 9 Logs

ClearPass Policy Manager / Audit Viewer